

Sentiment Analysis in Drug Reviews using Machine Learning and Deep Learning Techniques

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I. SPECIFICATION OF THE HYPOTHESIS

The rapid growth of social media and sites for customer feedback has resulted in a plethora of user-generated reviews, particularly medicine reviews. These reviews are useful to healthcare practitioners, pharmaceutical businesses, and regulatory authorities because they provide thorough information about the effectiveness and safety of a medicine. Drug review is required to improve medication usage and health outcomes by identifying and treating drug-related disorders. However, the massive number of assessments and specialized medical expertise required constitute a significant obstacle. Manually examining all of these comments is impracticable, making NLP activities such as sentiment analysis vital for dynamically extracting valuable information.

Hypothesis: “Through the application of advanced natural language processing (NLP) techniques, our goal is to build a robust model capable of properly predicting sentiment polarity in medication reviews. This model is expected to provide valuable insights on user perceptions regarding medication outcomes. We contend that the strength of feelings is closely linked to a variety of factors, including the frequency of reported side effects and the severity of the medical condition under treatment. Our hypothesis proposes a high correlation between the sentiment expressed in user-generated reviews about medical treatments and drugs, and the perceived efficacy and safety of these products. Positive sentiment is thought to be connected with feedback of efficacy and minimal side effects, whereas negative sentiment is supposed to be associated with expressions of ineffectiveness and negative side effects”.

II. CRITICAL ASSESSMENT OF APPLICABLE TECHNIQUES

Pre-processing techniques:

• Tokenization

It is a vital preprocessing step in NLP, dividing text into individual words or tokens. It enables sentiment analysis by analyzing sentiments associated with specific words or phrases. For drug reviews, it allows the exploration of sentiments regarding drug effectiveness, side effects, and dosage. [1].

• Stopwords removal

Stopwords are words that lack meaningful information or sentiment and can negatively impact accuracy and performance. Removing these stopwords during preprocessing is crucial for improving sentiment analysis tasks [2]. By eliminating stop words, noise is reduced, and sentiment analysis algorithms become more efficient.

• Stemming and Lemmatization

Stemming removes derivational suffixes and inflections, allowing words with the same stem to be treated as equivalent during retrieval. This improves search recall by ensuring word variations are accounted for. For example, both “running” and “runs” are stemmed to “run” [3].

Lemmatization takes word normalization further by returning words to their base or dictionary form. It considers vocabulary, morphology, and the context of word usage. It helps differentiate between noun and verb forms and utilizes synonym databases to expand search results. For instance, a search for “hot” would also include results for “warm” [3].

• Word embedding

It is fundamentally a form of word representation that links the human understanding of knowledge meaningfully to the understanding of a machine. It is scattered depiction of a text in an n-dimensional space, which tries to capture the word meanings [4].

• Handling Negative Adjectives

Negation is when words, phrases, or sentences express the opposite meaning. Negation can happen in different ways, like using negative words or phrases. To handle negation in a sentence, one simple approach is to mark all the words between a negation cue (like “not”) and the next punctuation as negated. For example, in the sentence “I do not like this movie”, the word “like” is negated because of the negation cue “not”. [5].

Model development:

In the context of NLP tasks, each technique in Table I is going to be considered for different purposes in model development:

Supervised learning using ML:

This approach involves using machine learning algorithms such as Naive Bayes, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). Actually, these machine learning

TABLE I: Techniques used in model development

Approach	ML/DL	Technique
Supervised learning	ML	Naive Bayes, Support Vector Machine (SVM), K-Nearest Neighbors (KNN)
Supervised learning	DL	LSTM
Unsupervised learning	ML	K-means

techniques have been use in various sentiment analysis studies before with the advantages of less complex interpretation and less training time. However, low accuracy is a drawback of these algorithms in NLP tasks [6] [7]. In this research, text reviews will be transformed by TF-IDF or 'countvectorizer' before being fed into the model for training.

Supervised learning using DL:

Deep Learning technique is employed in this approach, specifically using Long Short-Term Memory (LSTM) networks. This algorithms has been gaining popularity in NLP tasks in general, and sentiment analysis in particular. In previous studies, LSTM has been successful in sentiment tasks with greater accuracy than traditional techniques [8]. In contrast, because of black-box characteristics, it is still complicated for interpret, require computational resource for the task as well as take much time in training the model.

Unsupervised learning with K-means:

This unsupervised learning approach utilizes the K-means clustering algorithm to identify clusters of reviews with similar sentiment patterns. In fact, using clustering-based approach can provide insights for different sentiments associated with specific drugs or conditions that would assist in data-driven decision making. It was considered as an executed effectively, useful, and non-human participant approach when it comes to tackle sentiment analysis problems [9] [10].

III. IMPLEMENTATION OF THE CHOSEN TECHNIQUES

The Figure 1 depicts the pre-processing of the raw text data through the steps of data cleaning. It serves as the necessary step for further analysis and modeling in sentiment analysis tasks.

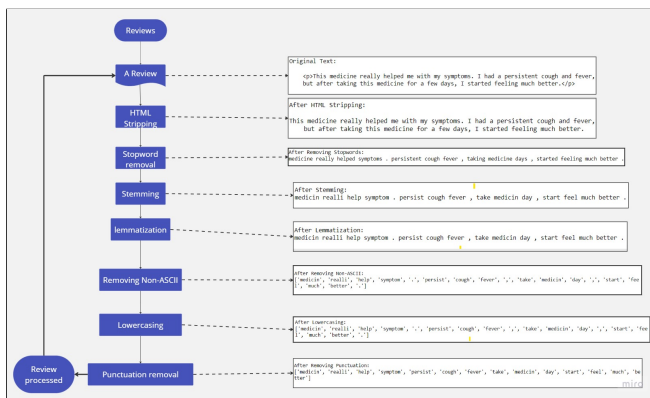


Fig. 1: Flow of pre-processing data

Figure 2 illustrates the process flow for NLP tasks in drug review analysis. The project employed both Supervised and Unsupervised Learning techniques to analyze sentiments in given reviews. The supervised approach involves employing LSTM, KNN, SVM, and Naive Bayes (NB) to train a model that accurately categorizes or predicts aspects of drug reviews [6] [7] [8]. These models can reveal the relationship between user emotions and ratings. Meanwhile, K-means clustering, an unsupervised technique, is used to group reviews based on language patterns and feelings, which is beneficial to data-driven decision-making [9] [10].

IV. RESULTS AND INTERPRETATION

A. Preprocessing and Exploratory Data Analysis

Figure 3 illustrates the rating's frequency, that varies from 1 to 10. Generally, there are more positive ratings than negative ratings. On the other hand, the positive rates are more spread into different classes, mostly between 8 and 10, while negative rates are concentrated into rating 1. This indicates that negative reviews are more intense than positive ratings.

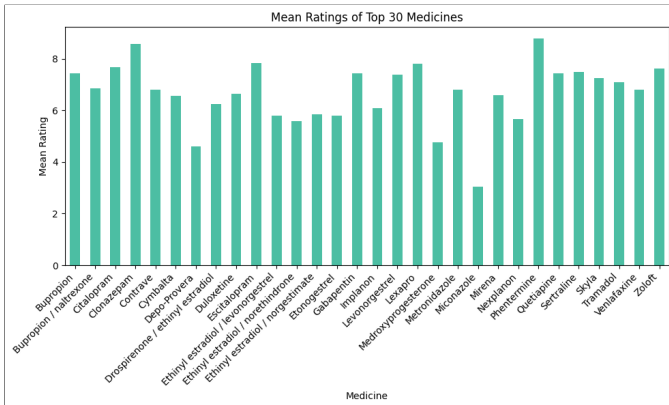
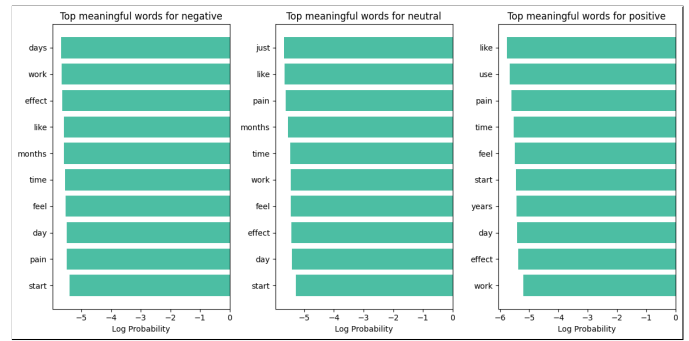
The Figure 4 displays the top 20 most frequently mentioned medical conditions in the reviews. Notably, 'birth control' emerges as the most frequently discussed among the medical conditions mentioned in the reviews, following birth control, the discussions in the reviews gravitate towards significant concerns such as depression, pain and anxiety.

The Figures 5 and 6, for example, stand out the most frequency word mentioned in reviews regarding diabetes and insomnia. Reviews about drugs related to diabetes emphasizes the use of phrases and words such as 'day,' 'side effect,' 'blood sugar,' 'A1C,' 'Victoza,' 'month,' or 'injection.' From insomnia reviews, the most common phrases and words are 'sleep,' 'night', 'hour', 'work', 'take', or 'Ambien'. These extensively displayed words allude to a focus on pharmaceutical effects and time constraints. Their existence indicates worries about the time required for a specific treatment to provide beneficial benefits. Although, other concerns are more specific for each condition.

Because one medicine can cure multiple ailments, the drugs are linked to at least one condition. According to customer ratings, the mean rating of top 30 drugs is shown in Figure 7. Specifically, medicines such as “Miconazole” and “Depo-Provera” have average mean ratings under 5. This could imply that consumers are dissatisfied with the outcomes or experiences they have had with these treatments. Medicines like “Phentermine” and “Clonazepam” stand out, with average mean scores above 8. These higher scores suggest that users have a more positive attitude toward these treatments, maybe as a result of their perceived effectiveness and beneficial effects.

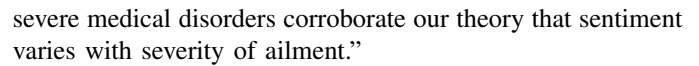
B. Supervised Learning

According to the performance scores shown in Figure 8, the LSTM model had the highest score (75%), followed by Naive Bayes (74%), SVM (66%), and KNN (57%). In consideration of choosing model for drug review sentiment analysis, the



with negative sentiments due to the frustration that comes with a failure to achieve desired results.

Figure 9 also depicts that the distribution of sentiment polarity in medication reviews can be connected with user's texts. Positive sentiment is common in reviews that reflect efficacy and few side effects, but negative sentiment is associated with ineffectiveness and unpleasant effects. The model output also stated that the findings support the proposed hypothesis, revealing a link between sentiment and perceived treatment effectiveness and safety. "Stronger feelings in reviews for



As can be seen in Figure 10, it becomes evident that positive sentiment predominantly surfaces in reviews reflecting effectiveness and minimal side effects, while negative sentiment is closely associated with perceived ineffectiveness and undesirable outcomes. The model's output reinforces the proposed hypothesis, elucidating a clear connection between sentiment and perceived efficacy and safety of treatments. Notably, the model's findings also underscore the correlation between heightened emotional expressions in reviews of severe medical disorders, lending further support to the theory that sentiment dynamics correspond to the severity of ailments.

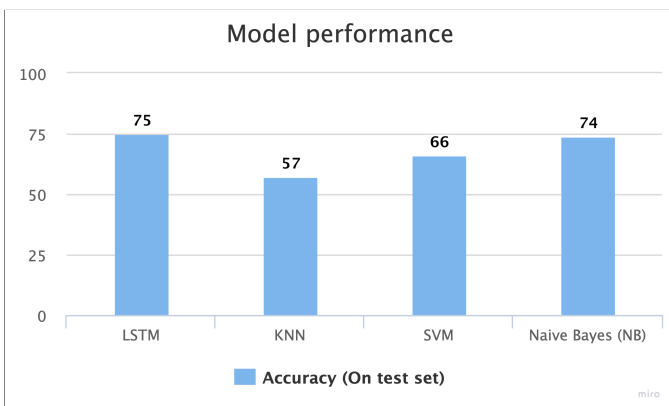
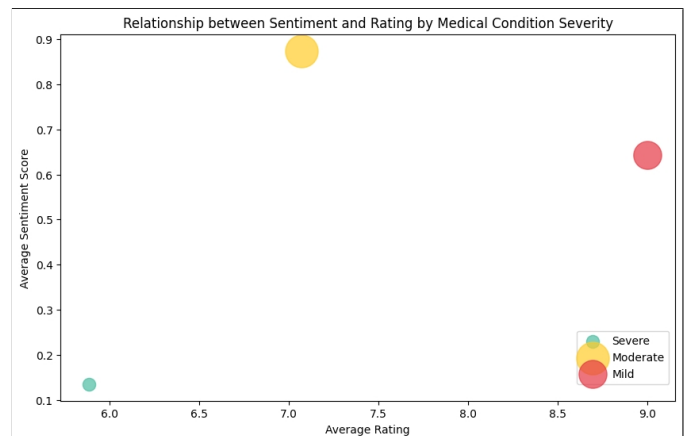


Fig. 10: Relationship between Sentiment and Rating by Medical Condition Severity

C. Unsupervised Learning

The clustering performed using the K-means technique Grouped the reviews into 8 different clusters. The 6 most important are represented in Figure 11 with the most common phrases of each.

In the Cluster 1, this cluster appears to be linked to discussions about skincare and acne treatment. The frequent use of phrases such as “acne,” “skin,” “month,” “use,” and “face” indicates that users are discussing their experiences with various treatments and techniques to enhance their skin



Fig. 11: Term frequency per cluster

condition. The prominence of phrases like as 'clear,' 'week,' and 'take' implies that people are sharing their progress and the efficacy of various skincare regimes.

Cluster 2 appears to be concentrated on pain management and alleviation conversations. The phrase 'pain' has the highest frequency, indicating that people in this cluster are most likely discussing their experiences with various pain relief therapies. Other often used phrases, such as 'take,' 'day,' and 'receive,' imply conversations regarding prescription schedules and effectiveness. The terms 'work' and 'impact' are also present, potentially suggesting how well medicines address users' pain difficulties. The prominence of phrases such as 'back' and 'year' may indicate that longer-term pain concerns are being mentioned.

Cluster 3 appears to focus around weight management and birth control discussions, regarding body well-being. The phrases 'weight' and 'gain' have the highest frequency, indicating that users in this cluster are discussing weight fluctuations, potentially owing to medication or other circumstances. The terms 'control' and 'birth' refer to discussions concerning birth control methods and their consequences. The occurrence of the words 'period' and 'pill' suggests that there have been discussions concerning menstrual cycles and birth control tablets.

Cluster 5 appears to be centered on comments about pharmaceutical effects and experiences. The phrase 'effect' appears the most frequently, indicating that people in this cluster are likely discussing the effects of various drugs. The terms 'take,' 'side,' 'medic,' and 'drug' all imply discussions about different aspects of pharmaceuticals, such as dose, potential side effects, and overall experiences. The terms 'year,' 'day,' 'get,' 'go,' and 'work' may refer to conversations concerning drug length, dose frequency, and the impact of pharmaceuticals on everyday life and activities.

Cluster 6 appears to be dominated by discussions about

mental health and emotional well-being. The phrase 'anxiety' is prominent, implying that users in this cluster are discussing their anxiety feelings and thoughts. The expressions 'take,' 'feel,' 'depress,' and 'day' refer to discussions about how drugs affect mood, emotions, and overall mental state. The terms 'go,' 'year,' 'effect,' 'get,' and 'start' may refer to talks regarding treatment duration, therapeutic effect onset, and overall impact on users' emotional and mental well-being.

It was shown that, in drug reviews, valuable information can be extracted to be applied by different business. The practical meaning and implications of the sentiment analysis model developed is precious for many stakeholders, including healthcare providers, pharmaceutical corporations, and regulatory authorities.

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